

Databases

Exercise Sheet 2

Klingemann, WS 2024 / 2025

1) Consider the following relations:

R:

A	B
a	b
c	b
d	e

S:

B	C
b	c
f	a
b	d

T1:

C
b
c

S1:

A	B
b	a
d	e
c	d

Calculate the following relations:

- $\pi_A(R)$
- $\pi_B(R)$
- $\pi_{A,B}(R)$
- $\sigma_{A=d}(R)$
- $\pi_A(\sigma_{B=b}(R))$
- $R \times S$
- $\sigma_{A=C}(R \times S)$
- $R \bowtie S$
- $\pi_A(R \bowtie S)$
- $\pi_A(R) \bowtie S$
- $R \bowtie T1$
- $R \bowtie S1$
- The theta-join of R and S with the join-condition $\Theta = R.B < C$.
We assume that $a < b < c < d < e < f$.

2) Consider the following schema:

Student(MatrnNr, SName, Birthday, Semester)

Formulate the following query in relational algebra

- What is the name and semester of the student with the MatrNr 4711?